Clearly Explained: How Machine learning is different from Data Mining

Definitions, confusions, distinctions- all explained

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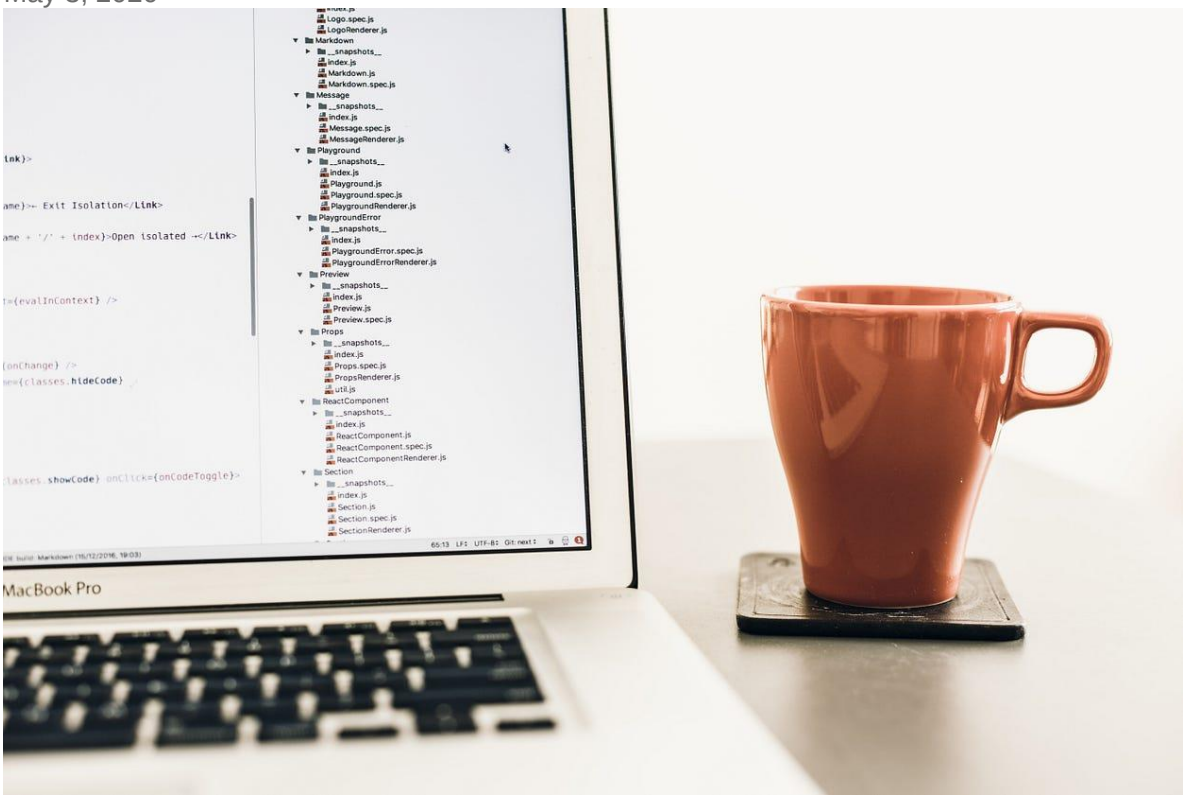

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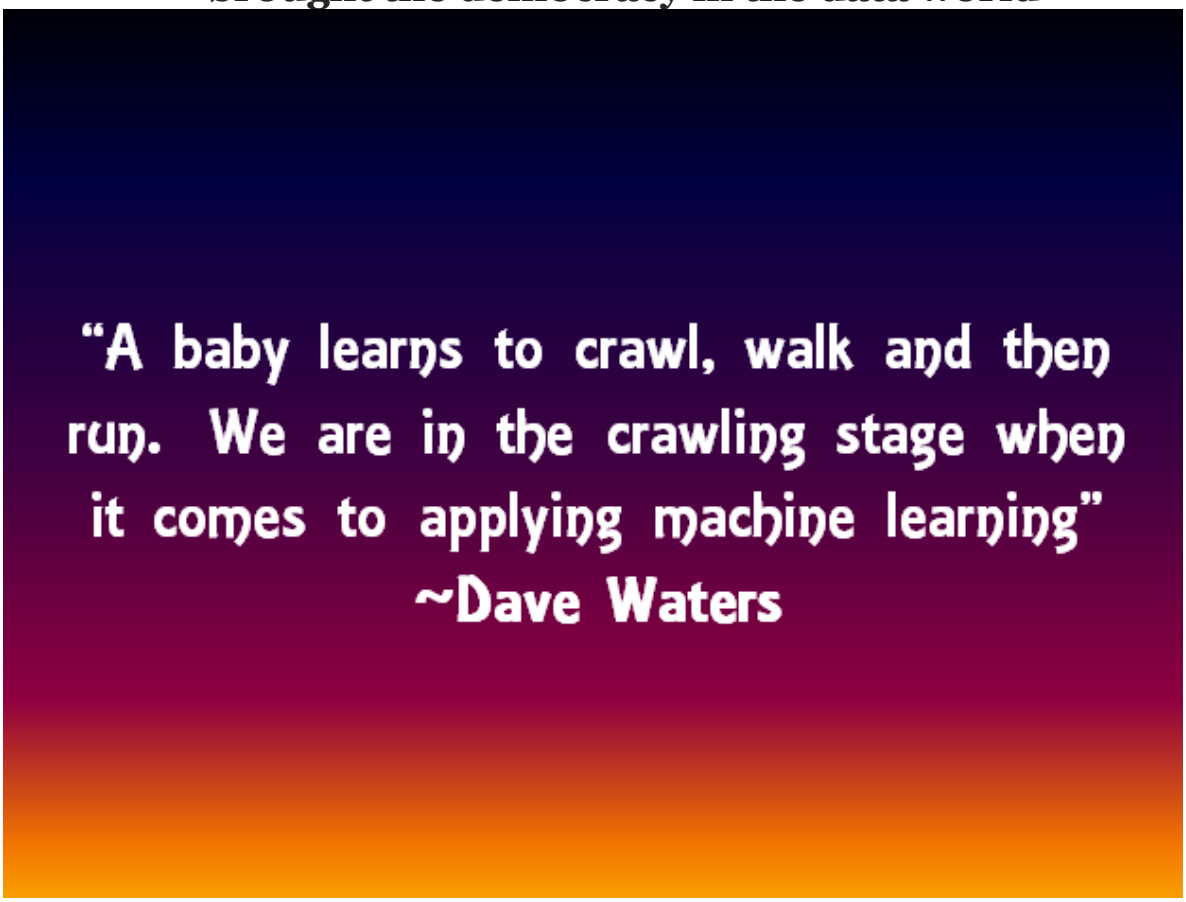
The flow of this post will be as follows:

1. **Talking about the era of big data and why ML has become so popular nowadays?**
2. **History and Points of confusion between the fields of data mining and Machine learning**
3. **Definitions of data mining and machine learning**
4. **Difference between data mining and machine learning**

Human beings start analyzing data since their birth. A child is exposed to a vast amount of signals (through eyes, ears, nose, tongue, and nerves) and his brain learns/adapts quickly as to how to analyze these signals and react to them. After the spoken language, then came the advent of written language which created vast repositories of data that can be analyzed to date.

So many people state that we have entered the ***era of Big Data*** just now, but the data was all around us always, it's just that the amount of *"Recorded"* data has started booming due to **4 main factors which have also contributed to Machine Learning becoming so popular nowadays** *(The following 4 points were put forth in an informative guide released by AnalyticsVidhya)*:

1. **Low storage costs**
2. **Low computational costs**
3. **Super-high number of devices that are connected to the internet now — ~50 billion connected devices by 2020**
4. **And of course, the cloud technology which has brought the democracy in the data world**



History

Now, let's talk a bit more about history to understand the two fields that this post is all about- **Data Mining and Machine Learning**

Data Mining came into being in the 1930s, originally known as knowledge discovery in databases, and Machine learning was introduced around the 1950s when the first ML program Samuel's Checker program was released.

Points of confusion between Data Mining and Machine Learning

1. Both are analytical processes and are an integral part of the Data Science industry.
2. Both came into being from their initial usages in pattern recognition use cases.
3. Both constitute learning from large amounts of data.

Definitions of Data Mining and Machine Learning

Data Mining

It involves a systematic hunt for nuggets of actionable intelligence in the existing data available.

Machine Learning

The field of study interested in the development of computer algorithms to transform the data into intelligent action is known as Machine learning.

Now, that the boring definitions are out of our way, let's jump into the most interesting part, the differences!

Differences between Data Mining & Machine Learning

Data Mining is a subset of business analytics and it focuses on teaching a computer — how to identify previously unknown patterns, relationships, or anomalies in the large data sets that humans can then use to solve a business problem. **This is a manual process that requires human intervention and decision making. The process can be made intelligent by a human by introducing smart features.**

For example: If a credit card company has a lot of data on customer behavior and they want to understand the patterns of customer attrition, then they will deploy data mining techniques to find out associations between different customer characteristics which can strongly indicate customer attrition in future.

WHEREAS

Machine Learning is a subset of Artificial Intelligence and it focuses on teaching a computer — how to learn to analyze large datasets and "learn" the patterns in it *(from the training dataset)* that can help make predictions on new data. **Apart from initial programming and fine-tuning, the computer doesn't need human intervention at all to learn from data and improve on its predictions by itself. The machine will become intelligent by itself as it feeds on more and more data.**

For example: Netflix is a perfect example of Machine learning. I saw Money Heist today on my suggestions list, so, how can Netflix figure out what I would like to watch next? There is a strong machine learning algorithm in place at the backend which is continuously analyzing the viewing preferences of other users with similar profiles as mine and hence it makes the suggestions based on that. As you watch more, the algorithm learns more about your preferences of the genre and you'll get the suggestions based on that!

Closing Notes

Virtually, all the Data Mining involves the use of Machine learning but not all Machine learning involves Data Mining.

Example: You might apply machine learning to data mine automobile traffic data for patterns related to accident rates

but on the hand if we talk about Self-driven cars, they are based on purely machine learning without any data mining being involved.

In continuation of this, you can read my following post to understand the 4 types of Machine learning algorithms! Do check it out!